

Belief or Action? Semantic Ambiguity in the Italian Non-finite Domain

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Abstract: This paper examines the belief/intention semantic alternation found in Italian clause-embedding verbs like *convincere* ('convince'), a phenomenon that is typically attributed to finiteness or mood (Giannakidou & Mari 2021, Grano 2018, 2024). Interestingly, Italian exhibits this alternation within its non-finite domain: *di*-infinitives convey belief readings, while *a*-infinitives give rise to intention readings. We argue that this semantic distinction corresponds directly to the structural size of the complement clauses. Syntactic evidence shows that *di*-infinitives project larger structures, accommodating modal auxiliaries and allowing both subject and object control, consistent with a propositional interpretation. In contrast, *a*-infinitives are structurally smaller, lack higher functional projections, and allow only object control. Our semantic analysis proposes that *a* selects complements whose eventuality argument remains unsaturated, thereby establishing a direct causal link to the matrix attitude, in line with previous intention-encoding accounts (Grano 2024), whereas *di*-infinitives, selecting larger structures, are fully propositional.

Keywords: *rational attitudes, belief/intention alternation, infinitival complements*

1 Introduction

Crosslinguistically, it has been observed that certain clause embedding verbs exhibit a belief/intention semantic alternation with respect to their complement clause (Giannakidou & Mari 2021, Grano 2018, 2024). Such alternation correlates with specific morpho-syntactic features of the complement clause, for example finiteness for English *persuade* (1) and Russian *dumat* (2), Indicative vs. Subjunctive mood for Greek *pitho* 'persuade' (3).

- (1) a. Mary persuaded John to leave.
b. Mary persuaded John that it was raining.

(English, Grano 2018, 2024)

- (2) a. Vasja dumaet vypit' piva.
 Vasja think-PRS.3SG drink-INF beer.GEN
 'Vasja intends to drink beer.'
- b. Vasja dumaet čto my idem pit' pivo.
 Vasja think-PRS.3SG that we go-PRS.1PL drink-INF beer.ACC
 'Vasja thinks that we are going to drink beer.'
- (Russian, Kasenov 2023)
- (3) a. I Ariadne epise ton Nikolas na fiji noris.
 the Ariadne persuade-PST.3SG the Nikolas SBJV leave-PRS.3SG early
 'Ariadne persuaded/convinced Nikolas to leave early.'
- b. I Ariadne epise ton Nikolas oti i idea tou ine kali.
 the Ariadne persuade-PST.3SG the Nikolas that.IND the idea his be.PRS.3SG good
 'Ariadne persuaded/convinced Nikolas that his idea is good.'
- (Modern Greek, Giannakidou & Mari 2021)

Interestingly, Italian *convincere* 'convince' (alongside other verbs, like *pensare* 'think', and *avere paura* 'fear'), shows the same alternation within the non-finite domain. The verb can select two infinitival prepositional complementizers:

- (4) a. Marco ha convinto Gianni **di** avere un figlio.
 Marco₁ have-PRS.3SG convince-PST.PTCP Gianni₂ *di* PRO_{1/2} have-INF a child
 'Marco has convinced Gianni that he has a child'
- b. Marco ha convinto Gianni **a** avere un figlio.
 Marco₁ have-PRS.3SG convince-PST.PTCP Gianni₂ *a* PRO₁ have-INF a child
 'Marco has convinced Gianni to have a child'

In (4a), Mario causes Gianni to have a belief about some state of affairs *s*, whereas in (4b) the result is that Gianni has an intention to bring *s* about. Moreover, only *di*-infinitives (4a) allow both subject and object control, while *a*-infinitives are only compatible with object control (4b). The two structures also differ in the temporal orientation of their complements, as *a*-infinitives are necessarily future-oriented (Wurmbrand 2014), i.e., they can only denote events that follow the reference time. As a result, they are ungrammatical with complement clauses describing a past event, contrary to *di*-infinitives (5). Note that, even in (4b), where no overt aspectual or tense morphology is present, the only available interpretation is that, at the time of the convincing, Gianni did not have a child yet.

- (5) a. Marco ha convinto Gianni **di** avere avuto un figlio dieci anni fa
 Marco₁ have-PRS.3SG convince-PST.PTCP Gianni₂ *di* have-INF have-PST.PTC a child ten years ago
 'Marco has convinced Gianni that he had a child ten years ago.'

- b. *Marco ha convinto Gianni ad avere avuto un
 Marco₁ have-PRS.3SG convince-PST.PTCP Gianni₂ a have-INF have-PST.PTC a
 figlio dieci anni fa.
 child ten years ago
 'Marco has convinced Gianni to have had a child ten years ago.'

In this paper, we explore the syntactic differences between these two infinitival structures and show how the interpretations of the two *convincere* reports are compositionally derived from a single lexical entry for *convincere* combined with each infinitival structure.

2 Previous accounts on the belief/intention alternation

In recent literature, belief/intention alternation has been discussed in the broader context of mood alternation. As illustrated in (3), the belief versus intention readings with Greek *pitho* correlate with mood choice (Indicative vs. Subjunctive) introduced by different complementizers (*oti* vs. *na*, respectively). In the framework of Giannakidou & Mari, Indicative selection presupposes subjective veridicality, namely that the embedded clause is evaluated as true relative to the modal background introduced by the embedding verb. Subjunctive selection, by contrast, signals a suppositional attitude with epistemic uncertainty regarding the truth of the complement.

Turning to English *persuade*, Grano (2018) synthesizes two influential ideas: first, that modal quantification in attitude reports is contributed by the complement clause rather than by the attitude verb itself (Bogal-Allbritten 2016, Kratzer 2006, 2013, Moulton 2009); second, that infinitives may carry built-in modal semantics (Bhatt 2008, Grano 2016). Leveraging these insights, Grano (2018) derives the alternation from the semantics of the complement: finite clauses introduce *information* modality, yielding a *cause-to-believe* reading, whereas (*for/to*) nonfinite clauses introduce *preference* modality, giving rise to a *cause-to-intend* reading.

- (6) a. $\llbracket \text{PRO to leave} \rrbracket = \lambda e. \forall w' \in \text{PREF}(e) : \exists e' [\text{Leave}(e') \ \& \ \text{Agent}(e, \text{PRO}) \text{ in } w']$
 b. $\llbracket \text{that it was raining} \rrbracket = \lambda e. \forall w' \in \text{INFO}(e) : \exists e' [\text{Rain}(e') \text{ in } w']$

On this account, both complements are eventuality descriptions, which allows *persuade* to maintain a single lexical entry in (7) targeting **rational attitudes** (a class including belief and intention, but not desire; see Grano 2012, 2018).

- (7) $\llbracket \text{persuade} \rrbracket^w = \lambda P_{vt}. \lambda x. \lambda y. \lambda e. \exists e'. \text{Persuade}(e) \wedge \text{Agent}(e, y, w) \wedge \text{Patient}(e, x, w) \wedge \text{CAUSE}(e, e') \wedge \text{RATIONAL-ATTITUDE}(e') \wedge \text{Experiencer}(x, e') \wedge P(e')$

According to (7), *persuade* takes in input an eventuality description and two individuals and returns an eventuality description: a persuasion event with agent y and patient x that causes an attitude eventuality e' with property P and experiencer x . Crucially, the attitude introduces the eventuality argument that serves as the modal anchor for the embedded quantification (Alonso-Ovalle, Menéndez-Benito & Rubinstein 2024, Hacquard 2010).

A limitation of this proposal, noted by Grano (2024), is that (6a) does not capture the satisfaction conditions of intention. As Searle (1983) observes, beliefs are satisfied by the truth of their content, whereas intentions require that the relevant action be causally dependent on the intention itself. To address this issue, Grano (2024) proposes that complements licensing intention readings—namely Subjunctive and non-finite clauses—abstract over their eventuality argument. This makes it possible to implement the requirement of **causal self-referentiality**: the state introduced by the attitude must causally contribute to the realization of the intended event described by the complement. This dependency is encoded in the semantics of *intend*, adapted in (8), where a CAUSE^* relation links the attitude state to the embedded event, and the complement supplies an abstracted eventuality argument available for binding.

$$(8) \quad \llbracket \text{intend} \rrbracket^w = \lambda P_{\langle e, \langle s, vt \rangle \rangle}. \lambda x_e. \lambda s_v. \exists s. \text{Intention}(s, w) \wedge \text{Experiencer}(s, x, w) \wedge \forall \langle w', y \rangle \in \text{CONTENT}(s) : \exists e. \text{CAUSE}^*(s, e, w') \wedge P(y)(w')(e)$$

Indicative mood, which aligns with belief readings, provides existential closure of the eventuality variable; as a result, Indicative clauses are unsuitable complements for *intend*. However, this mapping does not straightforwardly extend to Italian, where the belief/intention alternation often appears with two infinitival types rather than a finite versus non-finite contrast: *di*-infinitives pattern with belief readings even though they clearly do not combine with Indicative mood.

We therefore pursue an alternative solution mentioned by Grano (2024) (in turn based on a suggestion by Menéndez-Benito): closure of the eventuality argument is not effected by Indicative itself, but by a dedicated functional head located structurally above Subjunctive and below Indicative. Under this view, the semantic alternation reduces to a difference in clausal size. Accordingly, if independent evidence establishes that *di*- and *a*-infinitives differ in structural size, this mechanism can be extended straightforwardly to the non-finite domain, deriving the observed alternation from the interaction between complement size, eventuality abstraction, and the locus of modal quantification.

3 Evidence from structural size

In fact, the two infinitivals seem to exhibit crucial structural differences. To begin with, *di*-infinitives allow modal auxiliaries such as *potere* ('can') and *dovere* ('must') intervening between *di* and the infinitive, contrary to *a*-infinitives:

- (9) Marco ha convinto Gianni **di** voler / poter / saper /
 Marco have-PRS.3SG convince-PST.PTCP Gianni *di* want-INF / can-INF / be-able-INF /
 dover studiare.
 have-to-INF study-INF
 'Marco has convinced Gianni that he {wants to}/{can}/{knows how to}/{has to} study'

- (10) *Marco ha convinto Gianni a voler / poter / saper /
 Marco have-PRS.3SG convince-PST.PTCP Gianni *a* want-INF / can-INF / be-able-INF /
 dover studiare.
 have-to-INF study-INF
 ‘Marco has convinced Gianni to {wanting to}/{can}/{knowing how to}/{having to} study’

Since modal auxiliaries merge above the verbal domain (Cinque 2006, Grano 2012), their presence in *di*-infinitives but not in *a*-infinitives suggests that *di*-infinitives project a larger structure. This aligns with theoretical accounts linking structural size to propositionality (Lohninger & Wurmbrand 2020).

Moreover, we have already seen that only *di*-infinitives can exhibit both subject and object control, whereas *a*-infinitives only allow object control (as shown in (4b)/(4a)¹). This follows if we assume that the former represent a standard case of control infinitives, with the possibility of projecting up to FinP — the projection that hosts the logophoric center (cf. Bianchi 2003, Landau 2025, Rizzi 1997, a.o.) — and hence can mediate the control relationship between both the subject and the object. On the other hand, *a*-infinitives are expected to enforce object control if they lack this projection, given their reduced size: in such cases, the matrix object is simply the closest referent available for coindexation with the infinitival PRO.

Finally, only *di*-infinitives can be assessed for truth (11), a behaviour expected if they are propositional complements. By contrast, the fact that *a*-infinitives cannot be assessed for truth suggests that they are not structurally large enough to constitute propositions (cf. Wurmbrand & Lohninger 2023).

- (11) a. Marco ha convinto Gianni di avere un figlio, ma non
 Marco have-PRS.3SG convince-PST.PTCP Gianni DI have-INF a child, but not
 è vero.
 be-PRS.3SG true
 ‘Marco has convinced Gianni that he has a child, but it is not true.’

¹As generously pointed out to us by Carlo Geraci, there are cases in which a subject-control reading is available with *a*. A relevant example is provided below.

- (i) La modella ha convinto suo marito a farsi dipingere nuda.
 The model have-PRS.3SG convince-PST.PTCP her husband *a* make-INF=SE paint-INF naked.
 ‘The model has convinced her husband to be painted naked.’

We note that this reading has been judged somewhat marginal by most of the speakers we consulted. Our preliminary view is that in these cases, as the object is still the entity bearing the intention and agency required to carry out the infinitival event, the control relation might target the embedded agent/initiator introduced by the causative verb (Ramchand 2008), instead of the subject of the *portraying* event. We will discuss a similar mechanism below, when we will address the case of infinitival passive embedded by *a* (12). We leave a more detailed analysis of this kind of examples to future work.

- b. Marco ha convinto Gianni ad avere un figlio, # ma non
 Marco have-PRS.3SG convince-PST.PTCP Gianni A have-INF a child, but not
 è vero.
 be-PRS.3SG true
 ‘Marco has convinced Gianni to have a child, but it is not true.’

Momentarily leaving aside *di*-infinitives (as we assume they constitute a case of propositional infinitival complements), we will now focus on the syntactic and semantic properties of *a*-infinitives.

3.1 Zooming-in on *a*-infinitives

The reduced structure of *a*-infinitives raises the question of how a future interpretation is licensed (cf. (5)): in other words, if their structure lacks Tense, how do they force a future-oriented interpretation? To address this, we draw on the observation that root modality, located in the lower portion of the T layer, is inherently associated with future orientation (Kratzer 2013). This kind of modality is not restricted to low modal auxiliaries (Hacquard 2006, 2010): for instance, the English progressive has been proposed to introduce ‘inertia worlds’ (Copley 2009, Dowty 1979, Portner 1998), while (Wurmbrand 2014)), on the basis of (Abusch 2004), argues that future-oriented control structures involve a future-oriented modal operator (WOLL), situated above vP. Crucially, as argued by (Copley 2009), both the progressive and the future seem to share a similar modal component, involving a kind of inertial or bouletic ordering. In what follows, we show that *a*-infinitives provide evidence for precisely this kind of limited functional domain above vP, providing the space for a WOLL-like element encoding future-orientation.

Several pieces of syntactic evidence support the presence of structure above vP in *a*-infinitives. First, passive is available (12), as well as low aspectual restructuring verbs (13), which are merged at the edge of the vP domain (Cardinaletti & Shlonsky 2004, Cinque 2006, Grano 2012, Wurmbrand 2001).

- (12) La conduttrice ha convinto l’ospite ad essere
 the conductor have-PRS.3SG convince-PST.PTCP the guest TO be-INF
 intervistato.
 interviewed-PST.PRT.M
 ‘The conductor has convinced the guest to be interviewed.’
- (13) Marco ha convinto Gianni a cominciare a/continuare
 Marco have-PRS.3SG convince-PST.PTCP Gianni a begin-INF TO/continue-INF
 a/finire di lavorare.
 TO/finish-INF OF work-INF
 ‘Marco has convinced Gianni to start/continue/finish working.’

Interestingly, the infinitive in (12) necessarily implies an active (consenting) role in the interviewing event: Gianni does not passively enable it but rather allows it (it can be paraphrased as *Marco has convinced Gianni to let (someone) interview him*). This is consistent

with the idea that, with *a*-infinitives, the attitude holder (*Gianni* in (12)) has an intention of bringing about the event described in the complement clause.

Another indirect argument in favor of a low, circumstantial modal akin to *WOLL* resides in the lack of restructuring effects, despite the reduced nature of these complements. According to Cinque (2004, 2006), Wurmbrand (2001), and Wurmbrand & Lohninger (2023), transparency effects such as clitic climbing are evidence for the lack of a complete functional structure above the infinitival verb. Consider (14):

- (14) L'operaio lo_1 prova a costruire t_1 .
 the-worker it₁ try-PRS.3SG TO build 1
 'The worker tries to build it.'

The object clitic *lo* is able to climb over the matrix verb due to the fact that both the restructuring verb *tries* and the lexical verb *build* are in the same domain, and the infinitive is not higher than a vP. The case of *a*-infinitives embedded under the attitude verbs addressed in this paper, however, is different in that it does not allow clitic climbing.

- (15) *Gianni lo_1 pensa a costruire t_1 domani.
 Gianni it₁ think-PRS.3SG TO build t_1 tomorrow
 'Gianni thinks about building it tomorrow.'

Our structural hypothesis is further supported by the fact that these *a*-infinitives can host negation (16), suggesting that they must project some reduced functional structure above vP. This makes them comparable to the *situation* complements, as described by Wurmbrand & Lohninger (2023) and Lohninger & Wurmbrand (2020), separating them from restructuring infinitives which would be in fact *event* complements.

- (16) Gianni mi ha convinto a non salutare più il capo.
 Gianni me.CL have-PRS.3SG convince-PST.PTCP TO not greet-INF anymore the boss
 'Gianni has convinced me not to say hello to the new boss anymore.'

Further evidence for the presence of an underlying modal element comes from the distribution of temporal adverbs. Because *a*-infinitives constitute distinct temporal domains from the matrix clause, they permit independent temporal modification in both clauses (17). This contrasts with standard restructuring configurations, where the infinitival complement lacks independent temporal specification and is fully dependent on the temporal properties of the restructuring verb (18).

- (17) Ieri Gianni mi ha convinto a comprare una macchina
 yesterday Gianni me.CL have-PRS.3SG convince-PST.PTCP TO buy-INF a car
 nuova il mese prossimo.
 new the month next
 'Yesterday Gianni convinced me to buy a new car next month.'

- (18) *Ieri Gianni ha provato a riparare la macchina il mese
 Yesterday Gianni have-PRS.3SG try-PST.PTCP TO repair the car the month
 prossimo.
 next
 ‘Yesterday Gianni has tried to repair the car next month.’

The syntactic evidence is therefore compatible with the presence of a future-oriented modal operator, similar to *WOLL*, at the lower-edge of the T domain. We propose that this operator functions as a factual modal (see Kratzer 2013), with a circumstantial modal base, consisting of possibilities that share the relevant facts with the actual situation, and an inertial ordering source (Copley 2009, Dowty 1979, Portner 1998).

The fact that the modal basis is circumstantial and not epistemic correctly predicts that *convincere-a* reports would not be sensitive to *de dicto/de re* ambiguities. Consider a context in which Marco has told Gianni that Luca is the new boss, and that he should talk with him. Gianni believes Marco, and he goes to talk to him. However, Luca is not the new boss, and Marco told him a lie.

- (19) #Marco ha convinto Gianni a parlare con il nuovo capo
 Marco have-PRS.3SG convince-PST.PTCP Gianni TO speak-INF with the new boss
 ‘Marco convinced Gianni to speak with the new boss.’

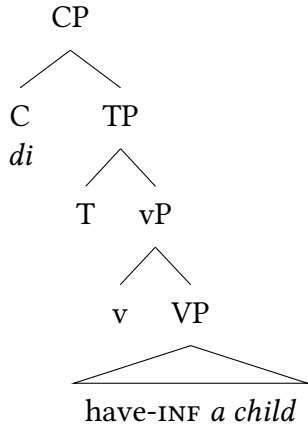
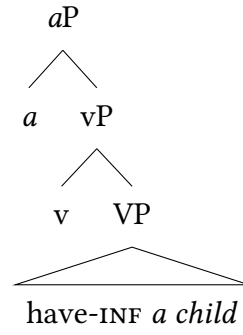
The unacceptability of (19) arises because when *a* is present, the definite description *the new boss* must refer to the entity in accordance with the *actual* circumstance, despite Gianni’s mistaken beliefs about who the new boss is. Note that *di-infinitive*, meaning *Marco has convinced Gianni that he is talking with the new boss* is perfectly fine with a *de dicto* reading of the definite description (arising from Gianni’s beliefs), consistent with its assumed propositional nature.

In the proposal laid out in the next section, we will assume for concreteness that the role of the future-oriented modal is played exactly by *a*. Following the grammaticalization pathway documented in Rice & Kabata (2007), we analyze *a* as retaining goal-directed semantics that yields future orientation in infinitival contexts. A piece of evidence in favor of this choice is that *a* is also involved in some forms of progressive which are available in some varieties of Italian:

- (20) Gianni sta a dormire.
 Gianni stays-PRS.3SG a sleep-INF
 ‘Gianni is sleeping.’

Assuming, as hinted above, that the progressive involves the projection of a set of ‘inertia-worlds’ (Copley 2009, Dowty 1979, Portner 1998), we can hypothesize that this modal component is provided exactly by *a*. However, we leave a more detailed analysis of this structure to future work.

Our exploration into the structural size of the two infinitival complements leads us to the following (sketched) representations.

(21) *di avere un figlio*(22) *ad avere un figlio*

4 Semantic analysis

The difference in size between the two infinitival structures has important semantic consequences (Lohninger & Wurmbrand 2020). We propose that *a* selects an infinitive whose eventuality argument remains unsaturated (Grano 2024), while *di* selects a larger, propositional structure in which the eventuality argument is saturated via a CLOSURE operator. This distinction is reflected in the denotations in (23).

- (23) a. $\llbracket [\text{vP PRO avere un figlio}] \rrbracket = \lambda e. \lambda w. \text{HAVE-A-CHILD}(e, w) \ \& \ \text{AGENT}(\text{PRO}, e, w)$
 b. $\llbracket [\text{TP CLOSURE PRO avere un figlio}] \rrbracket = \lambda w. \exists e'. \text{HAVE-A-CHILD}(e', w) \ \& \ \text{AGENT}(\text{PRO}, e', w)$

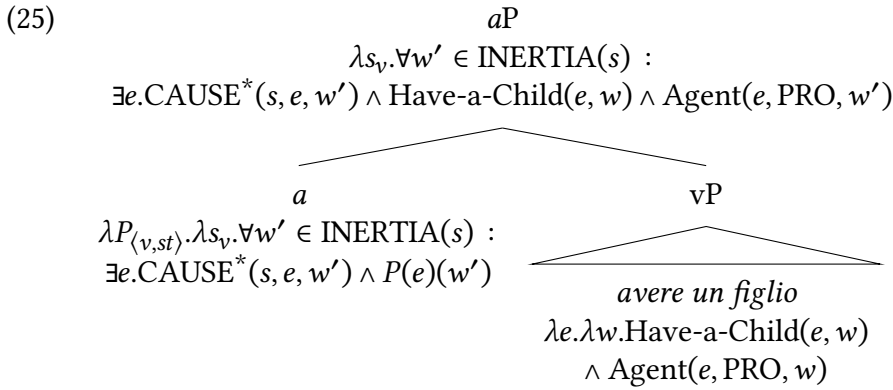
Following Grano (2024), intention encoding requires eventuality abstraction, so that the embedded event is causally linked to the matrix attitude. In the Italian non-finite domain, we argue that this causal link is introduced by *a*, yielding a ‘cause-to-intend’ reading, while *di* selects a proposition with an saturated eventuality argument. Assuming, with Kratzer (2006) and Moulton (2009), that modal quantification arises inside the embedded clause, we adopt a single denotation for *convincere*, building on Grano’s (2017, 2019) proposal for English *persuade*. On this view, *convincere* introduces a causal relation between an event of convincing and a resulting mental state classified as a rational attitude. As we saw in Section 2, rational attitudes include beliefs and intentions, but crucially not desires (Grano 2012, 2018). The denotation of *convincere* is given in (24).

- (24) $\llbracket \text{convincere} \rrbracket^w = \lambda P_{vt}. \lambda x. \lambda y. \lambda e. \exists e'. \text{Convince}(e) \wedge \text{Agent}(e, y, w) \wedge \text{Patient}(e, x, w) \wedge \text{CAUSE}(e, e') \wedge \text{RATIONAL-ATTITUDE}(e') \wedge \text{Experiencer}(x, e') \wedge P(e')$

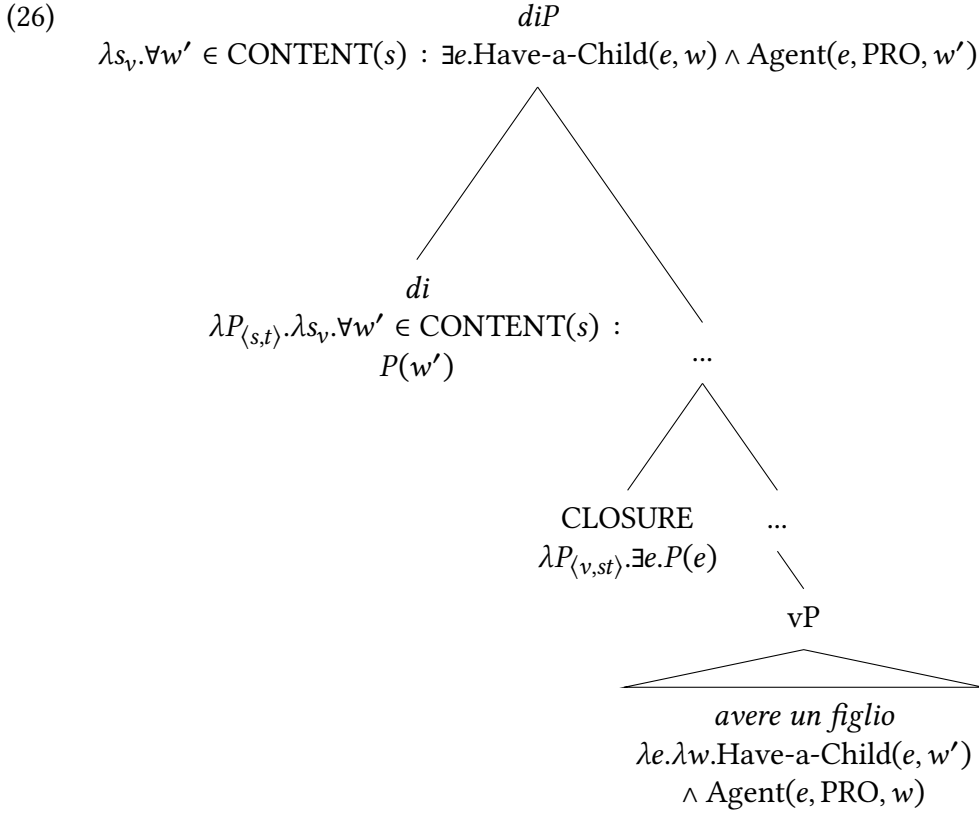
This lexical entry provides the core meaning of *convincere* independently of the type of infinitival complement. The differences between *a*- and *di*-infinitives emerge from the size of the complement and the way the eventuality argument is handled.

The bare vP infinitive in Italian denotes a proposition abstracting over world and eventuality arguments. When selected by *a*, this minimal structure receives additional functional

structure introducing a causal relation. Specifically, *a* introduces a CAUSE^* relation (Grano 2024) that binds the eventuality argument, thereby linking the intended event of the embedded clause to the attitude holder. This mechanism accounts for the causal self-referentiality of intention, namely, the intuition that the attitudinal state must causally contribute to the realization of the intended event. Additionally, *a* also introduces quantification over possible worlds, following Kratzer's (2006) and Moulton's (2009) insight that the attitude is located inside the complement clause, rather than in the attitude verb. Specifically, we propose that *a*, taking the attitude state as its modal anchor (Hacquard 2006, 2010), projects a set of INERTIA worlds derived from a circumstantial modal base and an inertial ordering source (see Section 3.1). The tree in (25) illustrates the composition for a case with an *a*-infinitive.



By contrast, *di* selects a larger clausal structure capable of hosting additional functional projections. The crucial difference is that the eventuality argument of the embedded clause is bound by an intervening CLOSURE operator. This operator existentially closes the event variable, yielding a propositional structure that abstracts only over the world argument. This is shown in (26).



In sum, both *a*- and *di*-infinitives yield properties of states, which can combine with the lexical meaning of *convincere*. The difference lies in whether the eventuality argument is and linked causally via *a*, or instead bound internally by a CLOSURE operator under *di*. This compositional distinction explains why both complements are compatible with a single lexical entry for *convincere*, while still producing the observed interpretive contrasts in the Italian infinitival domain.

5 Conclusions

This paper has investigated the belief/intention alternation in Italian embedded infinitives, with particular attention to the syntactic and semantic contrast between the two infinitival complements selected by verbs such as *convincere* ‘convince’. We have argued that, unlike in languages where the alternation correlates with finiteness or mood, Italian encodes the contrast through differences in the structural size of the complement. On the proposed analysis, *di*-infinitives correspond to larger clausal configurations, which can host additional functional structure responsible for eventuality closure and, consequently, a propositional interpretation. By contrast, *a*-infinitives instantiate a reduced structure that supports an intention reading via the introduction of a CAUSE* relation contributed by *a*. These structural asymmetries account for the observed interpretive contrast: *di*-infinitives yield belief-like readings, whereas *a*-infinitives give rise to intention readings characterized by causal self-referentiality.

Abbreviations

1 = first person, 2 = second person, 3 = third person, acc = accusative, cl = clitic pronoun, gen = genitive, inf = infinitive, ind = indicative, m = masculine, pl = plural, prs = present, pst = past, ptcp = participle, prt = participle, sg = singular, subjv = subjunctive.

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